ECON10005 Quantitative Methods 1

Final Tutorial in week 12: Week 11 Material

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Introduction

Zheng Fan

- Ph.D student in Economics at Unimelb.
- Personal website: *zhengfan.site* for some details

Don't be shy if you need help

- Discuss on Ed Discussion Board
- Attend consultation sessions: see Canvas for time and location
- Consult Stop 1, in case of special considerations,
- Contact QM-1@unimelb.edu.au for all admin issues, including group assignments.
- Send me an email: fan.z@unimelb.edu.au

Coefficient of Determination: R^2

"Goodness of Fit". Since $SST = SSR + SSE$, we can decompose R^2 as follows:

$$
\begin{aligned}
R^2 &= \frac{SSR}{SST} = 1 - \frac{SSE}{SST} \\
\text{(From Formula Sheet)} \qquad &= \frac{\sum_{i=1}^{n}(\hat{Y}_i - \bar{Y})^2}{\sum_{i=1}^{n}(Y_i - \bar{Y})^2} \\
&= \frac{\frac{1}{n-1}\sum_{i=1}^{n}(\hat{Y}_i - \bar{Y})^2}{\frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2} \\
&= \frac{\mathrm{Var}(\hat{Y})}{\mathrm{Var}(Y)}
\end{aligned}
$$

This variance ratio explains why you were able to quickly compute R^2 in last week's exercise.

Coefficient of Determination

"Goodness of Fit": R^2

The proportion of total variation in Y that is explained by the regression model (the independent variable X).

For a simple linear regression (with only one X variable), the sample correlation coefficient r is directly related to R^2:

$$r = \pm\sqrt{R^2}$$

(*Note: The sign of r matches the sign of the slope coefficient $\hat{\beta}_1$*).

Linear Regression: Statistical Inference

To test the hypothesis:

$$H_0 : \beta_1 = 0 \quad \text{vs.} \quad H_1 : \beta_1 \neq 0$$

We use the test statistic:

$$t = \frac{\hat{\beta}_1 - \beta_1}{\text{s.e.}(\hat{\beta}_1)} \sim t_{n-2}$$

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Where the standard error depends on the **Residual Variance** (s_U^2):

$$s_U^2 = \frac{1}{n-2} \sum_{i=1}^{n} \hat{U}_i^2 = \frac{SSE}{n-2} \quad (\text{with } df = n-2)$$

The resulting standard errors for the coefficients are:

$$\text{s.e.}(\hat{\beta}_1) = \frac{s_U}{\sqrt{(n-1)s_X^2}} \quad \text{and} \quad \text{s.e.}(\hat{\beta}_0) = \frac{s_U}{\sqrt{(n-1)s_X^2}} \cdot \sqrt{\frac{\sum_{i=1}^{n} X_i^2}{n}}$$

Tutorial questions

Suppose you have collected a random sample of 25 markets, with data on the average quantities of potatoes sold (Quantity Demanded) at various prices.

- The prices are given in **dollars per kilogram**, and the quantity demanded is measured in **tonnes**.

The second tab in the Excel file ‘Potato.xlsx’ contains data for these 25 randomly selected markets.

Tutorial questions

Assume a linear regression for the quantity demanded (QD) by this particular market as a function of price (P).

$$QD_i = \beta_0 + \beta_1 P_i + U_i$$

You estimated the sample regression line for this question last week. Using the results from this regression, we will now answer some questions.

Tutorial questions

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The estimated regression line is:

$$\widehat{QD}_i = 2.644 - 0.120 P_i$$

Can anyone remember from last week how we interpret the slope coefficient in the context of the demand function?

Tutorial question 1

The slope coefficient of -0.120 in the estimated regression equation represents the average change in quantity demanded (in tonnes) for a one-dollar increase in price.

Specifically, for each $1 increase in price, the quantity demanded decreases by 0.120 tonnes on average.

This negative relationship indicates that price and quantity demanded move in opposite directions, as expected for demand functions (following the law of demand).

Tutorial question 2

Conduct a hypothesis test at the 1% significance level to determine if there is a statistically significant linear relationship between the price of potatoes (P) and the quantity demanded (QD).

Let's spend 10-15 minutes working in groups on the whiteboards to answer this question.

Please use the probability tables to answer this question. And the estimates below.

	Coefficients	Standard Error
Intercept	2.644	0.017
Price	-0.120	0.007

Tutorial question 2

Understand that the question asks if the price is a significant determinant of the quantity demanded, so we will run a **two-tailed test** here.

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Hypotheses:

$$H_0 : \beta_1 = 0 \quad \text{Vs} \quad H_A : \beta_1 \neq 0$$

Decision Rule:

$$df = n - 2 = 23$$

$$\alpha = 0.01$$

The critical value is $t_{0.005,23} = 2.807$

i.e: Reject Ho if $|t| > t_{0.005,23} = 2.807$

Tutorial question 2

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Test Statistic:

$$t = \frac{\widehat{\beta}_1 - \beta_1}{SE\left(\widehat{\beta}_1\right)} \sim t_{n-2}$$

From the regression output:

$$\hat{\beta}_1 = -0.120; \quad SE\left(\hat{\beta}_1\right) = 0.007$$

Therefore:

$$t = \frac{-0.120 - 0}{0.007} = -17.14$$

Note: for a more accurate result, you should use more decimal places number for intermediate results. eg. use -0.11998 instead of -0.120.

As $17.14 > 2.807$, the decision is to reject the null hypothesis. Therefore, at the 5% significance level, there is sufficient evidence to suggest that price is a significant determinant of the quantity demanded.

Tutorial question 3

Construct a 99% confidence interval for the slope coefficient (β_1) in the regression model. Interpret what this interval tells you about the relationship between price and quantity demanded.

You have already constructed this confidence in the pre-tutorial. Let's review it together.

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$$\left(\hat{\beta}_1\right) \pm t_{0.005,23} SE\left(\hat{\beta}_1\right)$$

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$$(-0.120) \pm 2.807 \times 0.007$$
$$(-0.1396, -0.1004)$$

Based on this 99% confidence interval, is β_1 is significantly different from 0?

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Based on this 99% confidence interval, is β_1 is significantly different from 0?

At a 99% confidence level, this interval contains the true value of β_1. Because zero is not included in the interval, we can suggest that β_1 is significantly different from 0 at 1% level of significance.

Tutorial question 4

Elasticity of demand measures the responsiveness of quantity demanded to price changes. It is calculated as the percentage change in quantity demanded divided by the percentage change in price.

- If the absolute value of elasticity is greater than 1, demand is elastic (quantity demanded is highly responsive to price changes).
- If the absolute value of elasticity is less than 1, demand is inelastic (quantity demanded is less responsive to price changes).
- If the absolute value of elasticity equals 1, demand is said to be unitary.

Tutorial question 4

Using the coefficients from your regression equation, calculate the elasticity of demand at the price of \$2.104, which is the average price of the sample. Explain the elasticity of demand at this price point and interpret whether demand is elastic, inelastic, or unitary.

Let's spend 5-10 minutes working in groups to answer this question.

Hint: You should first estimate the quantity demanded at the price of \$2.104. The formula for the elasticity of demand (E_D) at a given price P and quantity demanded Q is:

$$E_D = \text{slope coefficient} \times \frac{P}{Q}$$

Tutorial question 4

Slope coefficient is -0.120.

P is the price = 2.104.

Q is the quantity demanded. Using the estimated regression:

$$\widehat{QD} = 2.644 - (0.120 \times 2.104) = 2.392$$

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Compute the elasticity of demand (E_D) at a given point:

$$\begin{aligned} E_D &= \text{slope coefficient} \times \frac{P}{Q} \\ &= -0.120 \times 2.104/2.392 = -0.106 \end{aligned}$$

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$|\text{Elasticity of Demand}| = |-0.106| = 0.106.$

Since the absolute value of elasticity is less than 1, demand is inelastic at this point.

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In other words, a 1% change in price would result in less than a 1% change in quantity demanded.

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Any questions for the tutorial?

Final Examination Details

- **Format:** Please note that the final exam will be a traditional **pen-and-paper-based** assessment (consistent with the format of your Mid-Semester Test).
- **Permitted Resources:** You will be provided with an formula sheet and probability tables during the exam (familiarise them before hand).
- **Canvas Resources:** Ensure that you carefully read all administrative instructions published in the dedicated *Final Exam Module* on Canvas.
- **Consultations:** I highly encourage you to attend the upcoming scheduled consultation sessions to address any remaining conceptual queries.

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- **Iterative Evaluation:** If you perform well, deepen your understanding of edge cases. If you struggle, cycle back to target and reinforce those specific weak points.

These tips are based on proven study practices. Please adapt them to the methods that work best for you.

Thank You and Good Luck!

Thank you for a wonderful semester!

Final Words of Encouragement

You have put in the work and mastered some complex econometric concepts. Trust your preparation, stay calm, and approach each problem step-by-step.

Best of luck with your final exams!